Electronics and Electrical Communications Engineering Department

Faculty of Engineering

Cairo University

Submitted to: Dr. Mohsen Rashwan

Neural Networks

Assignment 1
Classical Machine Learning Methods

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Prepared by:

Name	Section	ID
Yousef Khaled Omar Mahmoud	4	9220984
Salah	10	100
Shahd	10	100

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1 Assignment 1: Classical Machine Learning methods

This report is organized into three parts. Part 1 covers the regressing problems, Part 2 presents the classical machine learning modeling tasks on ReducedMNIST, and Part 3 covers the labelling-process pipelines on Indian_Digits_Train.

2 Part 1: Regressing Problems

This section is reserved for the regression problems from the assignment.

3 Part 2: some classical ML modeling

Part 2 uses the recovered ground-truth labels from `pipeline3/full_labels_vector.csv` and evaluates three feature representations: DCT, PCA, and HOG. Each feature set was tested with K-means per class using 1, 4, 16, and 32 clusters per class, and with SVM using linear and RBF kernels.

- **Best overall classifier:** HOG + RBF SVM with 99.80% accuracy.
- **Best K-means classifier:** HOG + 16 clusters per class with 99.40% accuracy.
- **PCA dimensionality:** 161 components were required to reach at least 95% explained variance.

Classifier	Spec	DCT_225		PCA_95		HOG	
		Accuracy (%)	Time (s)	Accuracy (%)	Time (s)	Accuracy (%)	Time (s)
K-means	$K = 1$	91.55	0.22	91.65	0.12	96.90	6.68
K-means	$K = 4$	95.90	0.65	95.95	0.63	98.65	14.98
K-means	$K = 16$	97.60	1.95	97.75	1.67	99.40	22.08
K-means	$K = 32$	97.90	3.47	97.75	2.53	99.35	33.54
SVM	Linear	98.15	0.72	97.85	0.59	99.70	4.61
SVM	RBF	98.55	1.85	98.95	2.05	99.80	11.69

Table 1: Full Part 2 comparative matrix required by the assignment

The best results are HOG + RBF SVM with 99.80% accuracy and HOG + K-means with $K = 16$ at 99.40% accuracy.

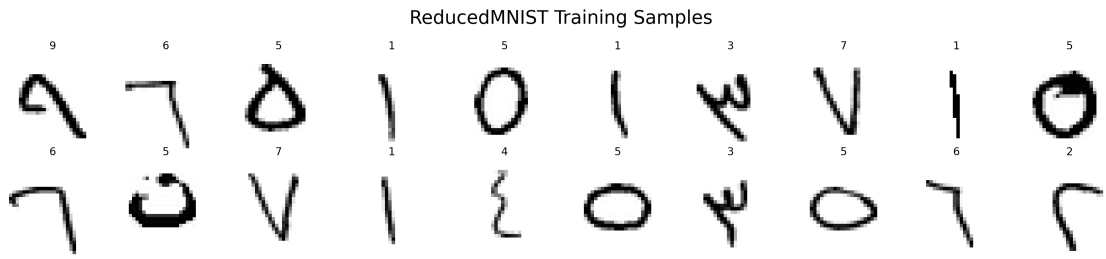

Figure 1: Sample images from the labeled dataset used in Part 2

This figure confirms that the dataset includes broad handwriting variability (stroke thickness, slant, and small shifts), which motivates using robust descriptors such as HOG and not relying only on raw pixels.

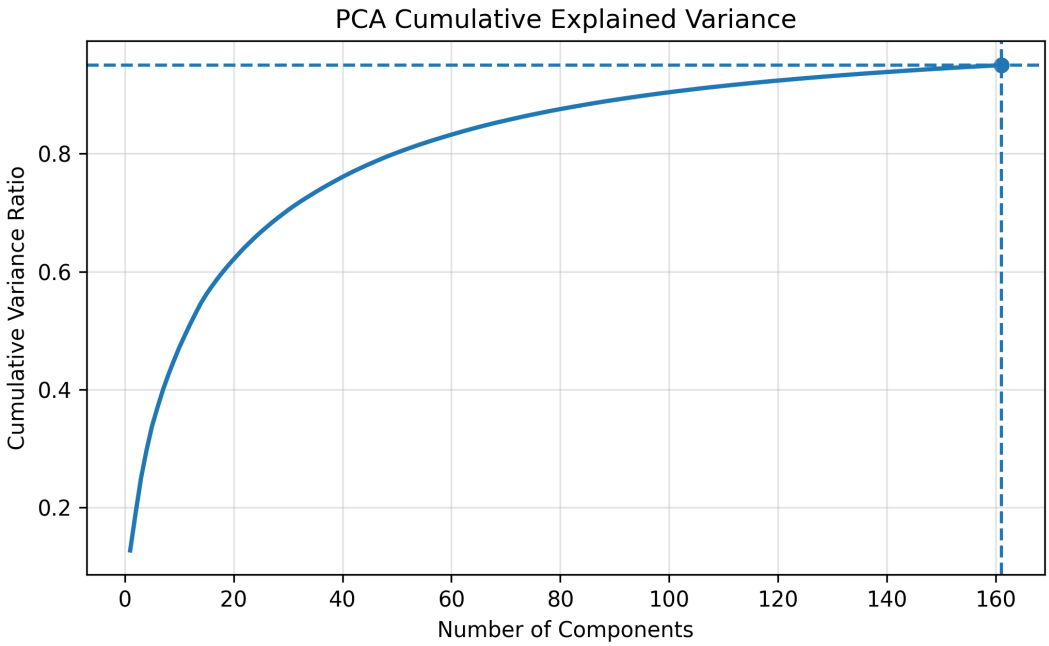

Figure 2: PCA cumulative explained variance curve

The PCA curve validates our component selection rule from the assignment. The retained dimensionality reaches the 95% variance target while still reducing feature size compared with the original 784-pixel representation.

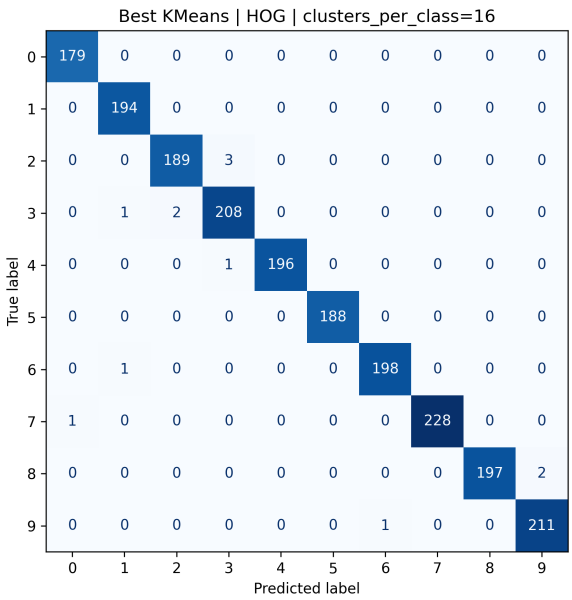

Figure 3: Best K-means confusion matrix for Part 2

The K-means confusion matrix shows that most predictions are concentrated on the di-

agonal, indicating strong cluster purity in the best configuration. Remaining off-diagonal cells highlight the visually similar digits that remain difficult under centroid-based classification.

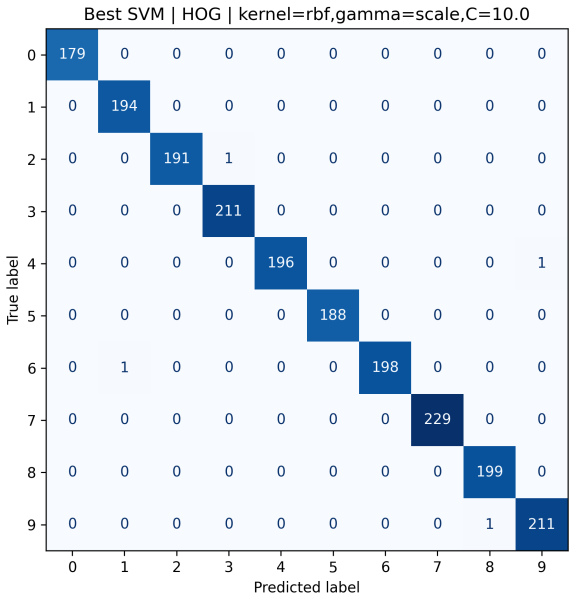

Figure 4: Best SVM confusion matrix for Part 2

Compared with K-means, the best SVM matrix has cleaner diagonal dominance and fewer confusion pockets. This directly supports the measured improvement in final accuracy for discriminative training over prototype matching.

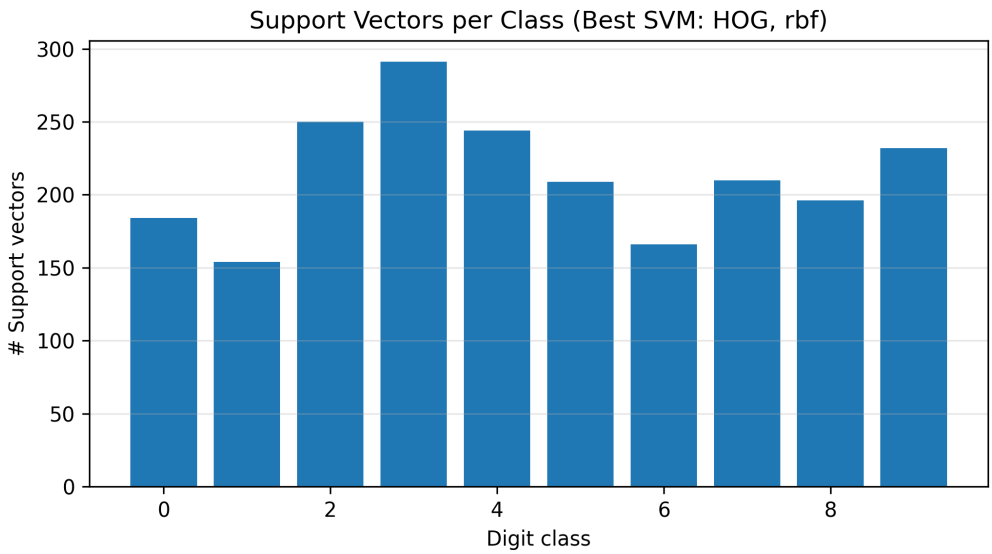

Figure 5: Support vectors per class for the best SVM setup

This figure summarizes model complexity by class. Classes with higher support-vector counts are closer to decision boundaries and require more boundary constraints, while lower-count classes are separated more easily.

The results show that HOG features provide the strongest representation for both clustering and classification. SVM consistently outperforms K-means, and the RBF kernel gives the best overall accuracy.

4 Part 3: ways to fasten the Labelling Process

5 Pipeline 1: K-Means Bootstrapping with Active SVM Refinement

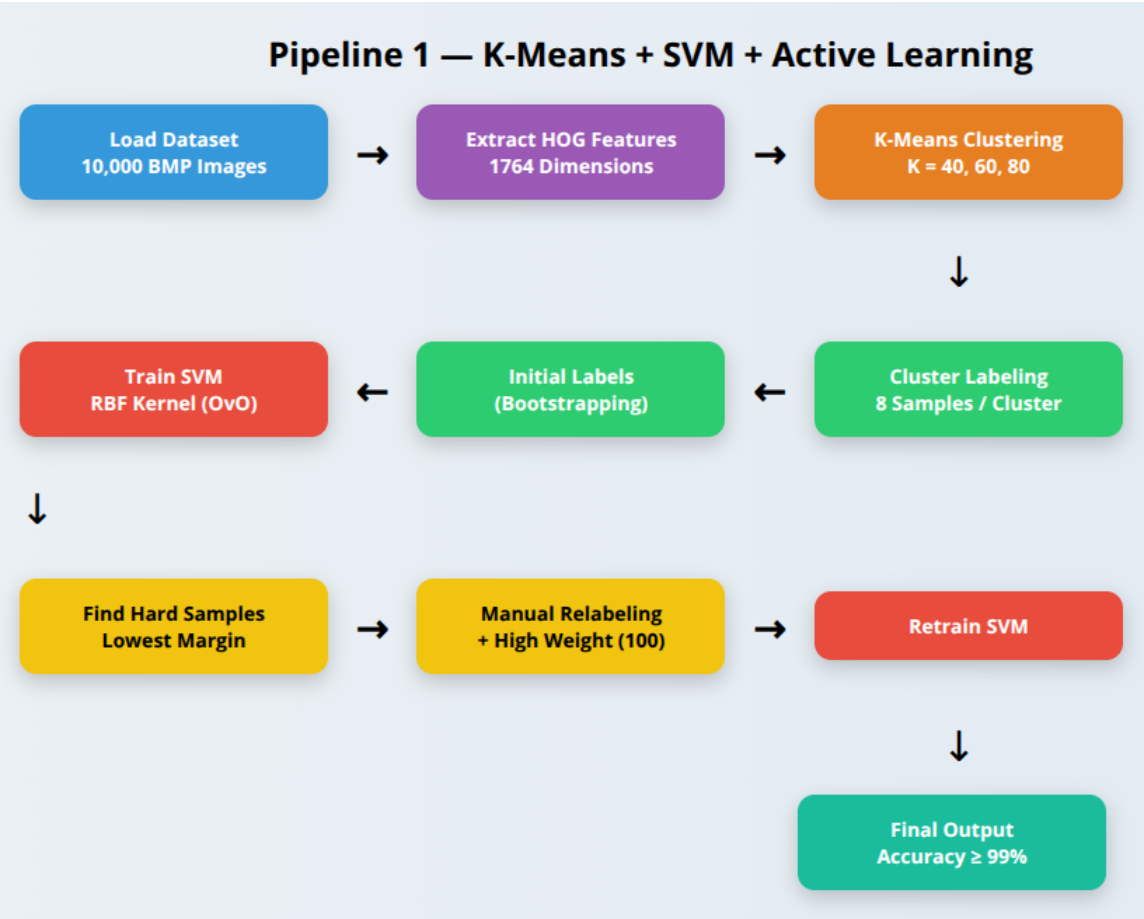

Figure 6: Pipeline 1: K-Means + SVM + Active Learning

The flowchart highlights the main design idea of Pipeline 1: use low-cost cluster-level supervision first, then spend manual effort only on uncertain boundary samples where human labels give maximum gain.

The pipeline consists of the following steps:

- Load dataset (10,000 BMP images)
- Extract HOG features
- Apply K-Means clustering
- Label clusters (bootstrapping)

- Train SVM classifier
- Apply active learning for refinement

6 Step 1 - Feature Extraction and K-Means Clustering

Histogram of Oriented Gradients (HOG) was used to extract features from images.

- Image size: 32 x 32
- Feature vector size: 1764

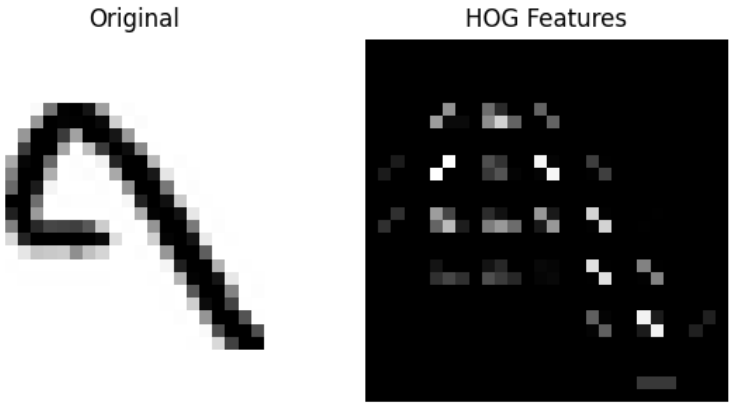

Figure 7: Example of HOG Feature Representation

The HOG visualization illustrates why this feature performs well in our experiments: it preserves local edge orientation patterns that are stable across many writing styles and minor geometric variations.

7 Step 2 - Human Labelling of Clusters (Bootstrapping Phase)

K-Means clustering was applied with different values of K:

- K = 40
- K = 60
- K = 80

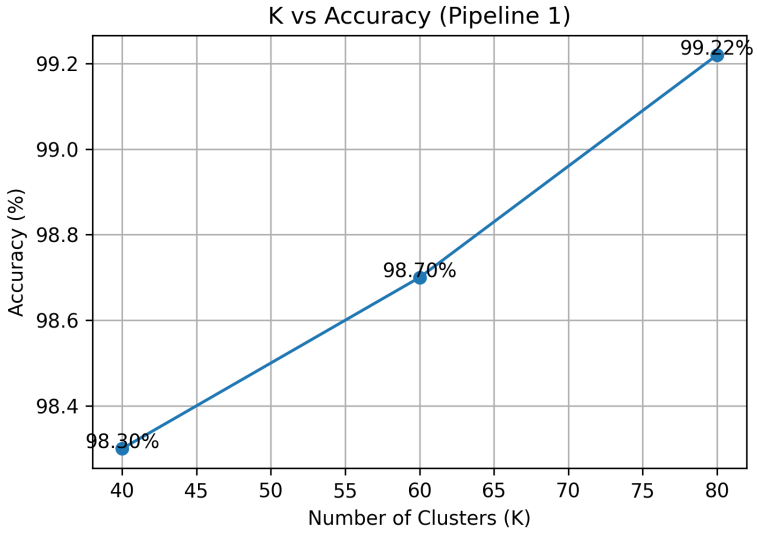

Figure 8: K vs Accuracy Before SVM

This plot shows that increasing K improves the initial pseudo-label quality up to the tested range, because smaller and more coherent clusters reduce mixed-class assignments during cluster bootstrapping.

8 Step 3 - Initial SVM Training and Step 4 - Active Refinement: Labelling Boundary Images

A multi-class SVM with RBF kernel (One-vs-One strategy) was used.

Active learning was applied as follows:

- Select 30 least confident samples (lowest margin)
- Relabel manually
- Assign weight = 100
- Retrain SVM

The process stops when:

- Accuracy $\geq 99\%$
- Improvement $< 0.1\%$

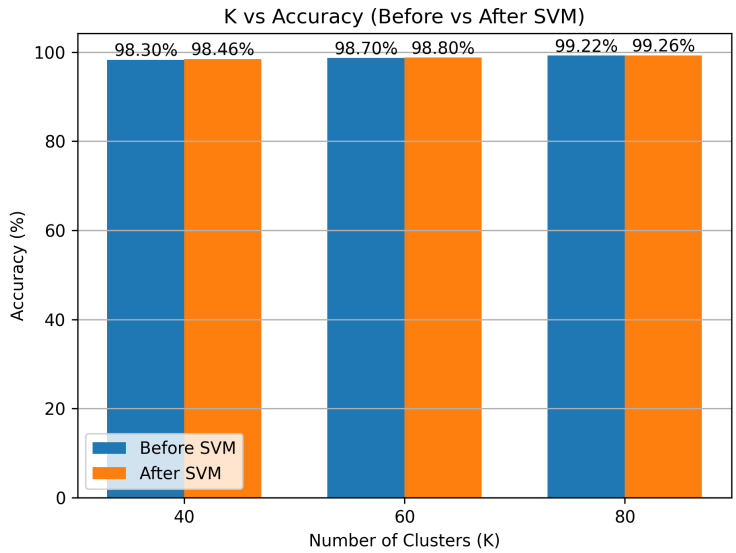

Figure 9: Accuracy Before vs After SVM

The bar chart quantifies the gain from SVM refinement over raw cluster labels. The gain is larger when initial clusters are weaker, and smaller when clustering is already highly pure.

9 Step 5 - Retrain and Evaluate

9.1 Accuracy

K	Before SVM (%)	After SVM (%)
40	98.30	98.46
60	98.70	98.80
80	99.22	99.26

Table 2: Accuracy Comparison

Table-based comparison confirms that the best operating point in Pipeline 1 is at high K with SVM refinement, reaching the required 99% threshold.

9.2 Iterations

K	Iterations
40	2
60	2
80	2

Table 3: Number of Iterations

The iteration table indicates fast convergence for all tested K values, which is consistent with using high-weight human corrections near the decision boundary.

10 Comparing Pipelines: Manual Labelling Time

For K = 80:

- Cluster labeling: 80 x 8 = 640 images
- Active learning: 30 x 2 = 60 images

Total images manually handled:

$$700 \text{ images}$$

Total time:

- Cluster labeling: 1600 seconds
- Active learning: 600 seconds

$$\text{Total time} = 2200 \text{ seconds} \approx 36.7 \text{ minutes}$$

11 Reporting Requirements

The proposed pipeline significantly reduces manual labeling effort from 27.8 hours to approximately 37 minutes while achieving 99.26% accuracy.

K-Means clustering provides strong initial labels, while SVM and active learning refine ambiguous samples. The improvement from SVM is more noticeable at lower K values, while higher K values already produce highly pure clusters.

12 Pipeline 2: Manual Seed + Augmentation + Self-Training

This pipeline adopts a semi-supervised learning approach starting from a small, high-quality human-labelled seed set. The model progressively expands its knowledge using data augmentation, active learning, and self-training.

12.1 Pipeline Diagram

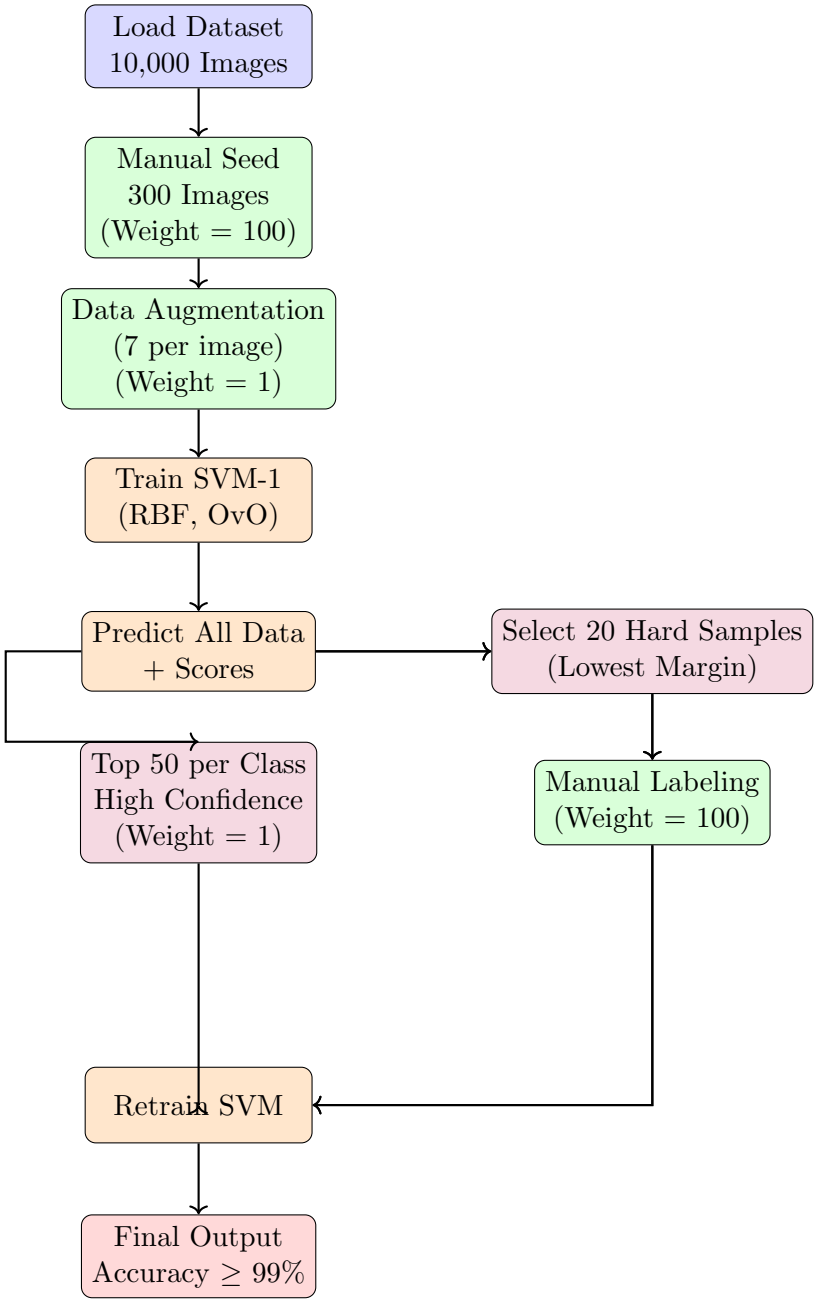

Figure 10: Pipeline 2: Semi-Supervised Learning Framework

The Pipeline 2 diagram emphasizes a different strategy: begin from a trusted manual seed, then expand supervision through augmentation, confidence-based pseudo-labeling, and iterative hard-sample correction.

12.2 Final Results

The final performance of Pipeline 2 after convergence is summarized as follows:

- Final Accuracy: 99.04%

- Number of Iterations: 4
- Total Manual Labels: 380
- Total Pseudo Labels: 1898
- Estimated Manual Time: 63.33 minutes

12.3 Iteration Progress

The improvement in model accuracy across iterations is shown in Figure below.

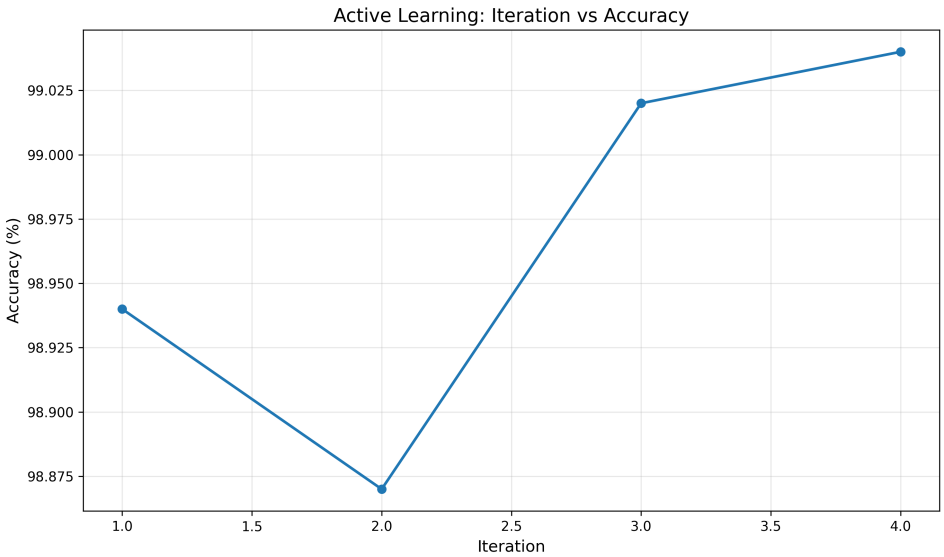

Figure 11: Active Learning: Iteration vs Accuracy

This curve demonstrates the expected diminishing returns of active learning: accuracy rises rapidly in early rounds, then improvements become smaller as the model approaches convergence.

12.4 Visualization

Figure 12: Initial Human Seed Samples (10 per class)

The seed gallery documents the initial trusted supervision used to anchor training. Quality at this stage is critical because early label noise can propagate through later pseudo-labeling.

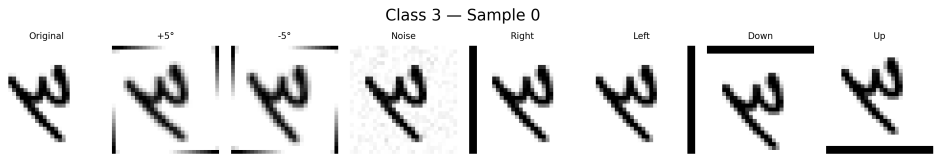

Figure 13: Example of Data Augmentation (7 variants)

The augmentation example shows the controlled transformations used to improve robustness. These variants help the classifier generalize beyond exact stroke shapes seen in the manually labeled seed.

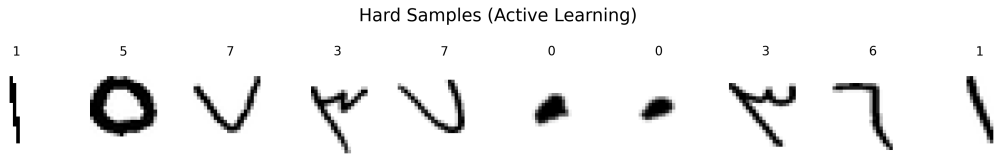

Figure 14: Hard Samples (Iteration 1)

Iteration 1 hard samples are primarily ambiguous boundary cases. Manually correcting these points has a high impact because they strongly influence decision surfaces.

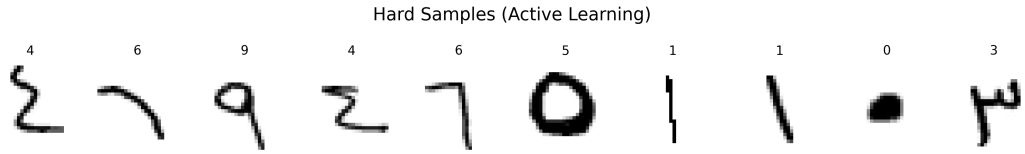

Figure 15: Hard Samples (Iteration 2)

By Iteration 2, hard samples become more specific and fewer broad ambiguities remain, which is consistent with progressive boundary tightening after each retraining cycle.

12.5 Manual Effort

Manual labeling consists of:

- Initial seed: 300 images
- Active learning: $20 \times 4 = 80$ images

Total manual labels:

$$300 + 80 = 380$$

Estimated time:

$$380 \times 10 = 3800 \text{ seconds} \approx 63.33 \text{ minutes}$$

12.6 Comparison with Pipeline 1

Metric	Pipeline 1	Pipeline 2
Accuracy	99.26%	99.04%
Manual Images	700	380
Manual Time	36.7 min	63.3 min
Iterations	2	4

Table 4: Pipeline Comparison

This comparison shows the practical trade-off between the two approaches: Pipeline 1 achieves slightly higher final accuracy, while Pipeline 2 reduces the number of manually labeled images by design.

Pipeline 1 achieves higher accuracy due to clustering-based initialization, while Pipeline 2 significantly reduces manual labeling effort by leveraging self-training. However, Pipeline 2 requires slightly more total time due to per-image labeling cost.

12.7 Discussion

Pipeline 2 demonstrates the effectiveness of semi-supervised learning. Starting from a small, high-quality seed set, the model successfully expands its training data using both active learning and self-training.

Although the final accuracy (99.04%) is slightly below Pipeline 1, the reduction in manual labeling effort is significant. This pipeline better reflects real-world scenarios where labeled data is scarce and expensive.

The combination of:

- Human supervision (high confidence)
- Model confidence (self-training)
- Iterative refinement

creates a scalable and efficient labeling strategy.

13 References

- Project Repository:
  `https://github.com/youefkh05/Neural_Netwok_projects`